Critical Region and Errors

The *critical region* of a test is that **part of the sample space** that corresponds to **rejection** of the null hypotheses H_0.
The *size* of a critical region, usually denoted by $\boldsymbol{\alpha}$, is the probability of sample point falling in the critical region when H_0 **is true**.

$$\alpha = P(X \in A_0 | H_0) \leftarrow \text{Type I Error}$$

A *best critical region* of size α is one that minimizes β.

$$\beta = P(X \in A_1 | H_1) \leftarrow \text{Type II Error}$$

A_0: critical region or rejection region

A_1: complement of critical region

$\boldsymbol{\alpha}$ is often known as **level of significance**

Calculation of A_0 , β for a given α

STEPS TO CALCULATE β

1. Decide an appropriate test statistic (usually this is a sufficient statistic or an estimator for the unknown parameter, whose distribution is known under H_0).
2. Determine the rejection region using a given α, and the distribution of the test statistic (TS).
3. Find the probability that the observed test statistic does not fall in the rejection region assuming H_a is true. This gives β. That is,

$$\beta = P(T.S. \text{ falls in the complement of the rejection region} | H_a \text{is true}).$$

A random sample of size 36 from a population with known variance, $\sigma^2 = 9$, yields a sample mean of $\bar{x} = 17$. Find β, for testing the hypothesis $H_0 : \mu = 15$ versus $H_a : \mu = 16$. Assume $\alpha = 0.05$.

Calculation of A_0 , β for a given α

Calculation of A_0 :

Here:

TS : $\overline{X}$

H_0 : μ = 15 vs H_a : μ = 16

Test Procedure: Reject H_0 if $\overline{X}$ > c where c is some constant to be determined.

$\alpha = P(\overline{X} > c | H_0)$ = 0.05. [given condition]

Distribution of TS: $\dfrac{\overline{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0, 1)$ [∵ n = 36 i.e. n is large]

Let Z = $\dfrac{\overline{X}-\mu_0}{\sigma/\sqrt{n}}$; Z ~ $N(0, 1)$; P(Z>$\dfrac{c-\mu_0}{\sigma/\sqrt{n}}$) = 1- Φ(c') = 0.05 where c' = $\dfrac{c-\mu_0}{\sigma/\sqrt{n}}$

Calculation of A_0 , β for a given α

Calculation of A_0 :

Let Z = $\frac{\overline{X}-\mu_0}{\sigma/\sqrt{n}}$; Z ~ $N(\ 0, 1\ )$; P(Z>$\frac{c-\mu_0}{\sigma/\sqrt{n}}$) = 1- Φ(c') = 0.05 where c' = $\frac{c-\mu_0}{\sigma/\sqrt{n}}$

From normal table : c' = 1.645

Hence c =μ_0 + 1.645 $\sigma/\sqrt{n}$ = 15.8225

A_0 = $\overline{x}$ ≥ 15.8225

Large Sample Approximation

Theorem 4.4.1 *Suppose that the population (not necessarily normal) from which samples are taken has a probability distribution with mean* μ *and variance* σ^2*. Then the standardized variable (or z-transform) associated with* $\overline{X}$*, given by*

$$Z = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}}$$

is asymptotically standard normal. That is,

$$\lim_{n\to\infty} P(Z \leq z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{z} e^{-u^2/2} du.$$

- ☞ above result follows from CLT.
- ☞ For practical purposes n $\geq$ 30.

Calculation of A_0, β for a given α

Then by definition,

$$\beta = P\,(\overline{X} \leq 15.8225 \text{ when } \mu = 16).$$

Consequently, for $\mu = 16$,

$$\beta = P\left(\frac{\overline{X} - 16}{\sigma/\sqrt{n}} \leq \frac{15.8225 - 16}{3/\sqrt{36}}\right)$$

$$= P\,(Z \leq -0.36)$$

$$= 0.3594.$$

That is, under the given information, there is a 35.94% chance of not rejecting a false null hypothesis.

Ack: Ramachandran et.al.

Trade off between α and β

Vintners and wine consumers continue to debate whether to seal wine bottles with corks or screw-tops. Screw-tops can reduce spoilage, but many wine enthusiasts associate them with cheap or otherwise undesirable wines. An often-cited 2011 survey by Rebecca Bleibaum (Tragon Corp.) found that about 20% of wine consumers would not buy screw-top wine, but negative attitudes toward screw-tops have abated over time.

A small winemaker will conduct a pilot study by surveying $n = 25$ randomly selected customers about their views on screw-top wine bottles. The parameter of interest is now p = the proportion of this winery's customers who refuse to buy screw-top wine.

What can be a possible test procedure? What can be possible hypotheses?

Ack: Devore et.al.

Trade off between α and β

Test statistic : X = the number of people who will not buy screw-top wine bottles

$H_0 : p = 0.2$ $H_a : p < 0.2$

More the parameters are apart lesser is β; i.e. chances are less that such a gap will go unnoticed

A_0, p_0, p_a	α	β
{ 0, 1, 2, 3 }, 0.2, 0.15	0.234	0.529
{ 0, 1, 2 }, 0.2, 0.15	0.098	0.746
{ 0, 1, 2, 3 }, 0.2, 0.1	0.234	0.236
{ 0, 1, 2 }, 0.2, 0.1	0.098	0.463

Suppose a study and a sample size are fixed and a test statistic is chosen. Then decreasing the size of the rejection region to obtain a smaller value of α results in a larger value of β for any particular parameter value consistent with H_a, and vice versa.

Power of a test

power = 1-β

x	0	1	2	3	4	5
$f(x \mid \theta_0)$	.02	.03	.05	.05	.35	.50
$f(x \mid \theta_1)$	.04	.05	.08	.12	.41	.30

A_0: critical region or rejection region

A_1: complement of critical region

$H_0 : \theta = \theta_0$. $H_1 : \theta = \theta_1$

Test procedure: Decide on the value of a single sample

$\alpha = 0.05$

Find A_0 and A_1.

Ack: Hoel, Port, Stone

Power of a test

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A_0: critical region or rejection region

A_1: complement of critical region

A_0	A_1	β
{ 0, 1 }	{ 2, 3, 4, 5 }	0.91
{ 2 }	{ 0, 1, 3, 4, 5 }	0.92
{ 3 }	{ 0, 1, 2, 4, 5 }	0.88

☜ α = 0.05

Best Test : A_0 = { 3 }

We can compare between tests keeping the significance level i.e. α fixed.